TASK: 1

Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

I installed Wireshark on my laptop and selected my active Wi-Fi network interface to capture network traffic. After starting the capture, I opened a website such as Instagram.com and stopped the capture after a few seconds. I observed the following protocols in the packet list:

Sr. No.	Protocol Name	Purpose
1	DNS	Converts domain names into IP addresses.
2	TCP	Provides reliable data transmission between devices.
3	TLS	Encrypts data to keep communication secure.
4	HTTP	Transfers web pages and other web resources.
5	QUIC	Supports fast and secure internet communication.
6	UDP	Transfers data without establishing a connection.
7	ARP	Helps identify the MAC address associated with an IPv4 address on a local network.

Conclusion: Wireshark helps us observe network traffic and understand how different protocols communicate when we access websites.

NETWORK

TASK: 2

**Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).

Hint: Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.**

Identifying the OSI Layers of Three Packet:

I selected three example packets from a Wireshark capture and identified their corresponding OSI layers.

Sr. No.	Protocol	OSI Layer	Explanation
1	DNS	Layer 7 – Application	DNS helps the browser find the IP address of a website.
2	TCP	Layer 4 – Transport	TCP provides reliable delivery of data between devices.
3	TLS	Layer 6 – Presentation (conceptually)	TLS encrypts and decrypts data to provide secure communication.

Conclusion: Different protocols work at different OSI layers to help data travel securely and efficiently across a network.

NETWORK

TASK: 3

Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

The OSI Model consists of seven layers. Each layer performs a specific function in network communication.

Layer No.	OSI Layer	Function	Real-World Example
7	Application	Provides network services to applications.	WhatsApp, Instagram, Chrome
6	Presentation	Handles data formatting, encryption, and encoding.	TLS encryption, JPEG images
5	Session	Establishes, manages, and terminates communication sessions.	Maintaining a communication session during a video call
4	Transport	Provides end-to-end data delivery.	TCP, UDP
3	Network	Routes packets between networks using IP addresses.	IP routing through a router
2	Data Link	Transfers frames within a local network using MAC addresses.	Ethernet, Wi-Fi
1	Physical	Transmits bits through physical or radio signals.	Ethernet cables, Wi-Fi radio signals

Diagram to draw:

Draw seven horizontal boxes stacked vertically. Give each box a different color and write the layer number, layer name, function, and example inside it.

NETWORK

THE OSI MODEL

7. APPLICATION WhatsApp, Instagram, Chrome
6. PRESENTATION Encryption, JPEG images
5. SESSION Communication sessions
4. TRANSPORT TCP, UDP
3. NETWORK IP addresses, routing
2. DATA LINK Ethernet, Wi-Fi, MAC
1. PHYSICAL Cables, radio signals

Conclusion: The OSI Model explains how data moves from an application through the network and reaches another device. Each layer performs a different task to make communication possible.

NETWORK

TASK: 4

Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Hint: Think about how your message is typed, packaged, sent over the internet, and received.

When I send a message on WhatsApp, the data passes through different networking layers before reaching the receiver.

1. **Application Layer (Layer 7):** I type a message in WhatsApp and press the Send button. The application prepares the message for transmission.
2. **Presentation Layer (Layer 6):** The data is represented in a suitable format and protected using encryption. WhatsApp uses end-to-end encryption for personal messages.
3. **Transport Layer (Layer 4):** Transport protocols help deliver the data between devices. Depending on the connection, WhatsApp may use TCP or other transport mechanisms.
4. **Network Layer (Layer 3):** The message data is carried in IP packets that contain source and destination IP addresses. Routers help forward these packets towards the destination.
5. **Data Link Layer (Layer 2):** The data is placed into frames for transmission over the local network, such as Wi-Fi. MAC addresses help deliver frames across the local link.
6. **Physical Layer (Layer 1):** The data is transmitted as radio signals over Wi-Fi or as electrical or optical signals through a wired network.

At the receiving end, the data passes through the layers in reverse order. WhatsApp processes the received message and makes it available to the recipient.

Conclusion: The OSI Model helps us understand how a WhatsApp message travels from the sender's device through the internet to the receiver's device. Each layer performs a specific function to support successful communication.